

Speaker-individuality in Fujisaki model f_0 features: implications for forensic voice comparison

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Abstract

Fundamental frequency (f_0) is a highly speaker-specific feature. Consequently, practitioners often use f_0 information in forensic casework. Current research principally examines the use of long-term f_0 statistics such as f_0 means and standard deviations for forensic voice comparison. The present study investigates how short-term f_0 features such as measured by the Fujisaki intonation model capture speaker-individuality. Based on data of a homogeneous group of Zurich German speakers, we conducted an experiment on a large corpus of read speech and on a subset of sentences that included speaking style variability (spontaneous vs read). The latter is characteristic of forensic casework. Speakers demonstrated high between-speaker variability and low within-speaker variability across the two speaking styles for a number of f_0 features. Given this evidence of speaker-individuality, we discuss Fujisaki f_0 features' potential for forensic voice comparison.

KEYWORDS SPEAKER-INDIVIDUALITY, PROSODY, INTONATION, SPEAKING STYLE
VARIABILITY

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1. Introduction

Intonation, or variability in voice fundamental frequency (f0), is a supra-segmental feature found in all languages of the world (Hirst and Di Christo 1998). The same way that punctuation helps to structure a written text, f0 is a tool for structuring information in spoken language. Variability in f0 is determined by various parameters: linguistically, f0 varies as a function of prominence marking (e.g., stressed vs unstressed) and sentence mode marking (e.g., declaratives vs questions). Further, it is used as a conversation-structuring device (Baumann 2006; Selting 1995). On a paralinguistic level, where the speaker deliberately adds information to modify the symbolic linguistic information (see Fujisaki 2004), emotion and attitude contribute to f0 variability (Kehrein 2002; Murray and Arnott 1993). On a non-linguistic level, i.e. on a level that is generally not controlled by the speaker (see Fujisaki 2004), sex, age, and the speaker himself contribute to variability in f0 (Bolinger 1989). In the present contribution we examine the degree to which the non-linguistic factor *speaker* contributes to variability in f0 features such as phrase declination (the global downtrend in f0 in an intonation phrase, hereafter IP) or the local excursion of f0 on particular syllables.

The phenomenon that non-linguistic factors such as *speaker* contribute to f0 variability is exploited in speaker identification and verification and in particular in forensic voice comparison (hereafter FVC, see Nolan 1983; Rose and Morrison 2009). In typical FVC cases, acoustic trace material from a crime (typically recordings of a perpetrator) is compared to acoustic comparison material (typically recordings of a suspect), and used for post-crime forensic investigations. Until today, forensic phonetic research has mainly revolved around f0 statistics such as means and standard deviations (see Braun 1992; Gold and French 2011; Jessen, Köster and Gfroerer 2005; Künzel, Masthoff and Köster 1995; Künzel 2000; Labutin, Koval and Raev 2007; Nolan 2002; Zellner Keller 2006) or baseline f0 values as proposed by Lindh and Eriksson (2007). There are presently only few studies that address how f0 features such as f0 declination or f0 alignment are speaker-specific and how this information could refine analyses of f0 for forensic purposes. Kraayeveld (1997), for instance, showed between-speaker differences in a number of f0 features such as f0 differences between the onset and end of the first rise in a declarative, the durations of the second rise, or the duration of final lowering in a declarative (terms as used by Kraayeveld 1997). Abberton and Fourcin (1978) reported that speakers could be identified by listeners on the basis of f0 information alone. They claimed that average f0 and f0 contour shape, i.e. range and rate of fall (terms as used by Abberton and Fourcin 1978), enabled such identification,

implying that these features are highly speaker-specific. In a tightly controlled experiment in which subjects were asked to reproduce the same intonation patterns as articulated by the experimenters, Nolan (2002) reported speaker-idiosyncrasies particularly in the points in time at which the f0 contour reached the f0 halfway (in Hertz) between the high and low f0 targets.

It is a basic tenet of FVC to accumulate as many characteristics of the speech signal as possible (see Nolan 1983; Rose and Morrison 2009). We believe that the previously studied f0 parameters do not exhaust the possible choice of parameters that may contribute to maximal speaker discrimination. It seems plausible that a more in-depth analysis of other fine-grained f0 features bears further potential for capturing speaker-individuality.

What exactly do we mean by more fine-grained f0 features?

1. f0 declination (the global downtrend in f0 to reach a minimum in phrase-final position)
2. IP duration
3. Local f0 excursions
4. Duration of local f0 excursions
5. Temporal alignment of f0 rise and fall relative to syllable onsets and offsets.

A majority of these f0 features have been successfully applied to reveal between-dialect differences (see Leemann 2012; O'Reilly 2013) and between-language differences (see Mixdorff 2004). To examine the potential of these f0 features for speaker-individuality, it appears useful to employ a model of f0 from which we can derive these parameters. We believe that the Fujisaki model (Fujisaki and Hirose 1982) allows for just that.

Conceptually, the Fujisaki model understands the f0 contour as a superposition of global and local components. From these two components, the above-mentioned f0 features can be derived (for a detailed explanation of the model and its derived parameters, see 2.4). The choice of the model is motivated as follows:

1. The Fujisaki model allows for quantification of f0 events and it does so by a fully automatic f0 contour parameterisation, which follows a manual pre-processing of the f0 contour (see 2.5). This semi-automatic, replicable procedure is advantageous if one wants to test for between-speaker differences in f0: all speakers are modelled according to the same principles. This increases the likelihood of finding differences that are attributable to the factor *speaker* without being a by-product of human modeller inconsistency. This aspect is critical bearing in mind the forensic context of the current study.

2. This semi-automatic procedure allows a larger amount of data to be handled than if human modellers were involved.

In the present study we examined both read speech and spontaneous speech in order to study within-speaker variability. These speaking styles were selected because forensic criminal and suspect recordings may exhibit a mismatch between the two: criminal recordings are most often spontaneous, while suspect recordings can be read and spontaneous speech. Given this possible mismatch it is important to know if the factor *speaking style* has an influence on the parameters being examined, which are the Fujisaki model f_0 features in our case. Ideally, parameters used in FVC show high between-speaker variability and low within-speaker variability (Nolan 1983). The type of speech material used in the present study – read and spontaneous speech – is thus of direct forensic relevance. Previous studies have revealed that, compared to spontaneous speech, read speech generally shows higher f_0 means (Braun 1995; Gfroerer and Wagner 1995; Hollien and Jackson 1973 – though there are many exceptions to this trend, especially in studies based on German, see Jessen 2009), greater correlation between the placement of IP boundaries and syntactic constituents (Botinis, Granström and Möbius 2001), greater alignment of pauses – which in turn may lead to pitch resets – with syntactic constituents (Duez 1982), and steeper f_0 declination (Umeda 1982; Vaissière 1983).

Given this evidence, we would expect to find differences between read and spontaneous speech in the Fujisaki model f_0 features examined. We addressed the following research questions:

1. Are there between-speaker differences in Fujisaki model f_0 features?
2. Are these possible between-speaker differences affected by speaking style?
3. What are the possible implications of (1) and (2) for forensic voice comparison?

2. Material and methods

2.1 Speakers and recording procedures

Sixteen speakers (8 male and 8 female) of Zurich Swiss German were recorded at the Phonetics Laboratory at the University of Zurich. The recordings were conducted in Swiss German by the first and fourth author. The speakers demonstrated little regional accent variability, as attested by informal listening tests. All speakers were natives of the variety of Zurich city German. The average age was 27 (SD = 3.6, range = 20–33). None of the speakers reported speech or hearing problems. The data was recorded in a sound-treated booth

using a Neumann STH-100 transducer microphone (sampling rate of 44.1kHz; 16 bit quantisation).

2.2 Material

2.2.1 *Spontaneous speech*

In the first recording session, spontaneous speech material was collected via semi-structured interviews. The 16 speakers were asked to talk freely to the interviewer (first and fourth author) about their studies at the University of Zurich. A subset of sentences, 16 per speaker (typically 15–20 syllables per sentence), was isolated from these interviews. We selected sentences according to prosodic, syntactic, pausing and breathing criteria. The isolated sentences had to form meaningful units and be fluently spoken. These 256 sentences (16 speakers $\times$ 16 sentences) formed the spontaneous corpus of this study.

2.2.2 *Read speech*

We constructed orthographic transcripts of these 256 spontaneous sentences (see Appendix for a sample of the recording material). These transcripts were handed out to the same 16 speakers with the request to practise reading the sentences for a second recording session. The 16 speakers read those 256 sentences at the University of Zurich (16 previously self-produced sentences + 240 sentences from their peers). These 4096 sentences (256 sentences $\times$ 16 speakers) constituted the read corpus of this study. The spontaneous and read corpora include 56,794 syllables in total.

2.3 Data annotation and processing

Trained phoneticians (first and fourth author) labelled the segmental onsets and offsets in the read and spontaneous speech corpora in Praat (Boersma and Weenink 2014) and transcribed each segment using SAMPA phonetic alphabet. The labellers cross-checked their segmentation on a regular basis in order to maximise inter-labeller consistency. Syllables were automatically annotated according to sonority hierarchy principles (Sievers 1881), where sonority decreases from the syllable nucleus to the syllable edges. Following this decrease, sonority starts to increase again, which is the locus where a syllable boundary was placed. Syllabification in Swiss German does not strictly proceed according to morpheme boundaries (Fleischer and Schmid 2006; Siebenhaar 2014). Syllable onsets are maximised before syllabification proceeds to the coda and nucleus of the preceding syllable (cf. Hall 2011). The sonority-based automatic syllabification was checked manually. This syllabification then served as a basis

for subsequent linking of the Fujisaki accent component timing parameters, T_1 and T_2 , to the relevant syllabic landmarks (onset and offset, respectively) (see Section 2.4)

Processing the data for subsequent Fujisaki modelling involved a number of steps. Initially, we created a pitch object in Praat (Boersma and Weenink 2014). The f0 contour was extracted based on the autocorrelation method presented in Boersma (1993). The frame duration was set at 10 ms, i.e. 100 pitch values were computed per second; pitch floor was set at 75 Hz. The f0 ceiling value for males was set at 300 Hz, for females at 600 Hz. We performed a manual check of the f0 contour following this automatic extraction (first author). Before the Fujisaki model was applied on the data, the extracted f0 contour was further processed using Praat's smoothing function (smoothing bandwidth: 10Hz).

2.4 Fujisaki model

The Fujisaki model represents f0 contours as a superposition of global and local contours (see 2.5 for the application of the model in the present study). In the f0 analysis process, the global and local components, which are called phrase and accent components, are added onto the baseline frequency F_b . The mathematical formulation of the model describes the resulting f0 contour $\ln F_0(t)$ as the sum of the baseline f0 and the phrase and accent components; see Equation 1.

Equation 1: Mathematical formulation of the Fujisaki model (see Fujisaki 1992b)

$$\ln F_0(t) = \ln F_b + \sum_{i=1}^I A p_i G_p(t - T_{0i}) + \sum_{j=1}^J A a_j [G_a(t - T_{1j}) - G_a(t - T_{2j})]$$

$$G_p(t) = \begin{cases} a^2 t \exp(-\alpha t), & \text{for } t \geq 0, \\ 0, & \text{for } t < 0. \end{cases}$$

$$G_a(t) = \begin{cases} \min[1 - (1 + \beta t) \exp(-\beta t), \gamma], & \text{for } t \geq 0, \\ 0, & \text{for } t < 0. \end{cases}$$

F_b : baseline value of fundamental frequency

I : number of phrase components

J : number of accent components

$A p_i$: magnitude of the i^{th} phrase component

$A a_j$: amplitude of the j^{th} accent component

T_{0i} : onset time of the i^{th} phrase component

T_{1j} : onset time of the j^{th} accent component

T_{2j} : offset time of the j^{th} accent component

α : natural angular frequency of the phrase control mechanism

β : natural angular frequency of the accent control mechanism

γ : relative ceiling level of accent components

$G_{pi}(t)$: i^{th} phrase control mechanism

$G_{aj}(t)$: j^{th} accent control mechanism

In the first formulation of the model, Fujisaki and Nagashima (1969) reported that the f0 contours in their data contained considerable variation between speakers. The authors proposed the use of a logarithmic scale to make the contours appear more similar. With the logarithmic domain, between-speaker differences as well as between-sex differences could be reduced. A decade later, Fujisaki, Tatsumi and Higuchi (1980) justify the logarithmic scale somewhat differently: the use of this scale allegedly reflects the stress–strain relationship of the human vocalis muscle. The logarithmic domain is said to mirror the linear relationship between muscle tension and muscle stiffness.

The three major model components – baseline f0, the phrase component and the accent component – are described in detail here with reference to Figure 1 below.

(1) Baseline f0 has seen different definitions over the years. In 1979 and 1981, Fujisaki defines F_b as ‘the lowest frequency of the vocal cord vibration’ (Fujisaki, Hirose and Ohta 1979: 166) and as ‘the lower limit of fundamental frequency below which vocal fold vibration cannot be sustained in the glottis of a speaker’ (Fujisaki 1981: 3). In 1987, Fujisaki (1987: 167) defines F_b as the ‘asymptotic value of fundamental frequency in the absence of accent components’, i.e. a more utterance-specific definition of F_b .

(2) The phrase component is the second addend in $\ln F_0(t)$. It models the slow, global changes in f0 and includes the following parameters: Ap , the impulse’s magnitude, T_0 , its onset time, and α , the time constant of the input signal that determines the degree to which the phrase contour is damped. All phrase component parameters, Ap , T_0 and α , are shown in Figure 1. Linguistically, the phrase commands correspond to intonation phrases. The phrase component is a crude approximation of declination, where an increase in the magnitude of the phrase component results in an increase in the slope of the declination (see Fujisaki and Hirose 1982). Perceptually, this increase in declination translates to an increase in pitch range, i.e. the variation in pitch between f0 maxima and f0 minima (Ladd 1996; Taylor 1994). The phrase component magnitude has been reported to be subject to phrase length, articulation rate, as well as the position of the impulse in the utterance (see Adriaens 1991; Fujisaki 1987; Fujisaki and Hirose 1982; Vaissière 1983).

(3) The accent component is the third addend in $\ln F_0(t)$. It includes the parameters T_1 , T_2 , Aa and β . Aa , accent component amplitude, models the size of a local f0 excursion, β determines the steepness of the rise and fall of an accent. T_1 and T_2 are the timing parameters of the accent component: T_1 is the point at which the local f0 contour starts to rise, T_2 is the point in time where the local f0 contour begins to fall. The accent component captures the local, quick f0 movements that are superposed onto the phrase component. All

accent component parameters are shown in Figure 1. Linguistically, the accent component has been associated with different prosodic units, including prosodic words (Fujisaki 1987), stress groups (Möbius 1993) or intonemes (Mixdorff 1998). Perceptually, accent commands are responsible for prominence marking at the local level, where prominence is expressed by increasing the amplitude of the component (Aa) (see Fujisaki 1992a). The timing of local accent commands is critical since it reflects the phonological nature of the accent type. Mixdorff and Fujisaki (2000) showed that different phonological pitch accents – as captured by ToBI labels (see Silverman, Beckman, Pitrelli, Ostendorf, Wightman, Price, Pierrehumbert and Hirschberg 1992) – can be identified by the relative onset times (T_1 -syllable onset) of the Fujisaki model accent component. Phonologically distinct accent types such as H^*+L and L^*+H can differ in terms of where – relative to syllable onset – the f_0 starts to rise. In H^*+L accents the rise is typically before syllable onset, while for L^*+H accents, the rise often starts in the middle of the corresponding syllable (see Figure 1). Based on these Fujisaki model parameters, we derive the following six model f_0 features used in the present study (for an illustration of these derived features, see Figure 1):

1. **Ap :** f_0 declination. The greater the Ap , the steeper the declination.
2. **$PC\ dur$:** duration of phrase component. Expressed as the interval duration between two T_0 s. The longer the interval duration, the longer the $PC\ dur$.
3. **Aa :** local excursion of f_0 . The greater Aa , the more pronounced the f_0 excursion.
4. **$AC\ dur$:** duration of local f_0 excursion. Measured as the interval duration of T_2-T_1 . The longer the interval duration, the longer the duration of the local rise-fall f_0 movement.

f_0 alignment is expressed relative to the segmental string. The way an f_0 rise, for example, is anchored with the segmental string, has implications for its phonetic and phonological interpretation. For this reason, the points where f_0 begins to rise or starts to fall, i.e. anchor points, are measured relative to syllable onset or syllable offset:

5. **$t1abs$:** temporal alignment of f_0 rise relative to syllable onset. Measured as the absolute temporal distance between T_1 and syllable onset only for those syllables that feature an f_0 rise as captured by the Fujisaki model (see 2.5). The earlier T_1 is relative to syllable onset, the smaller $t1abs$.
6. **$t2abs$:** temporal alignment of f_0 fall relative to syllable offset. Measured as the absolute temporal distance between T_2 and syllable offset only for those syllables that feature an f_0 fall as captured by the Fujisaki model (see 2.5). The earlier T_2 is relative to syllable offset, the smaller $t2abs$.

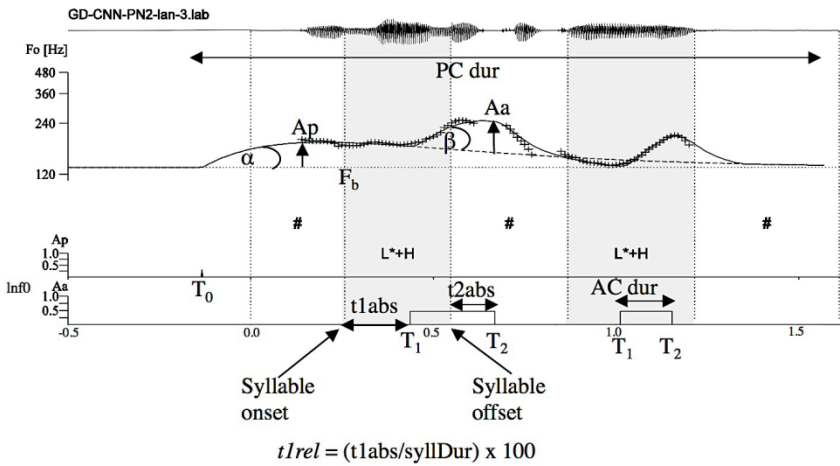


Figure 1: Illustration of the Fujisaki model f0 features. The top panel shows the speech waveform, the +++ demonstrate the extracted f0 values. The lower panel depicts the phrase component parameters A_p and T_0 . The accent components are shown as rectangular shapes (from T_1 to T_2) along with the accent component amplitude A_a . The boundaries of the accented syllables are marked with vertical dotted lines and the tonal transcription (here the L*+H labels) for each accent is provided.

2.5 Modelling procedure

The Fujisaki model parameters were configured as follows:

F_b : In the present approach we opted for a speaker-specific F_b . Keeping the baseline frequency constant per speaker will prevent variability in this parameter, thus allowing variability in the six derived Fujisaki model f0 features under investigation (see further below), which in turn should reflect speaker-specific characteristics. F_b is thus the speaker-specific, lowest voice fundamental frequency value onto which the phrase and accent components are superposed. In Figure 1, F_b is illustrated as a dotted line onto which the global phrase components and local accent components are superposed. F_b was established for every speaker by locating the lowest f0 measurement in a subset of 50 sentences per speaker in Praat (Boersma and Weenink 2014). The extracted f0 contours served as the basis for this procedure (for f0 extraction and subsequent manual check, see 2.3). The majority of f0 minima were found in utterance-final or utterance-medial positions. Particular attention was paid to passages of vocal fry (creaky voice), given that f0 measurements in these passages tend to be inaccurate. Inaccurate measurements were disregarded from this analysis. Once this lowest, speaker-

specific value of these 50 sentences was established, this value was then used for the subsequent Fujisaki modelling for every sentence of the speaker.

α : α was held constant at 2.0/sec (see Fujisaki et al. 1979; Mixdorff 1998; Leemann 2012).

β : β was held constant at 20.0/sec (see Mixdorff 1998; O'Reilly, Ní Chasaide and Gobl 2008; Leemann 2012).

The remaining model parameters A_p , A_a , T_0 , T_1 and T_2 , as well as the derived f_0 features $PC\ dur$, $AC\ dur$, $t1abs$ and $t2abs$, vary according to linguistic, paralinguistic and extralinguistic factors and may convey speaker-specific Fujisaki model f_0 information.

These model parameters were extracted with Mixdorff's *FujiParaEditor* (2009). The program estimated the Fujisaki model parameters based on the extracted f_0 contour. The parameter extraction is fully based on the f_0 contour which was previously corrected manually for micro-prosodic perturbations (see 2.3). The automatic parameter extraction is based on a cubed spline contour with which the extracted f_0 points are approximated. This continually smoothed contour is split into the high (HFC) and low (LFC) frequency components through a filter. These parts are used for the initialisation of the accent and phrase components. In HFC, accent commands are expected between two local minima, while in LFC, phrase commands are also expected at local minima. Based on these initial values, the local optimisation takes place iteratively (see Mixdorff 2000, 2009). The f_0 contour is automatically modelled to provide an optimal fit.

Given that the applied approach is automatic, the phrase component does not necessarily match phonologically determined intonation phrases. Rather, the automatic modelling procedure described below governs the assignment of the phrase command start and end points. Since in the current study we apply speaker-specific F_0 , it is possible that speakers do not always reach their lowest pitch in non-final phrases. In these instances, then, A_p does not necessarily reflect declination. Secondly, the accent component in the automatic modelling may not always reflect actual pitch accents unlike in a linguistically driven approach, where the accent component is associated with a prosodic unit (e.g., prosodic word (Fujisaki 1987), stress group (Möbius 1993) or intonemes (Mixdorff 1998)) and manual adjustment of commands is permitted. In the present study, however, accent commands are automatically assigned in such a way that they optimally capture local f_0 movements. The onset and offset of the accent command may thus coincide with either metrically strong or weak syllables; they therefore capture 'current' syllables, i.e. syllables in which T_1 , or T_2 , of the automatic fit is found.

2.6 Complementary ToBI analysis

Nolan (2002) reported that speakers can avail themselves of relative freedom as to how they align f0 events, which means that f0 alignment bears the potential to carry speaker-individual information. When all speakers read the same sentences but differ in f0 alignment values, these values may differ for two reasons: (1) speakers may have individual preferences for different pitch accent types (which in the intonational analysis such as ToBI would obtain a distinct label). The difference in the choice of accent type at the categorical level (such as L*+H versus (L+)H* which are of direct interest to us as they contain a rising onglide) translates in the Fujisaki model to late versus early timing of T_i relative to the syllable onset (i.e., *t1abs*, see Section 2.4). (2) Speakers may in fact use the same accent types but exhibit minor, possibly speaker-individual, differences in f0 alignment (see Nolan 2002).

To test for possible underlying causes for between-speaker differences in f0 alignment, we selected two exemplary speakers that strongly differ in *t1rel* values (see 3.1 for an explanation of *t1rel*). We performed an autosegmental-metrical analysis (see Goldsmith 1976) for a subset of 32 of the 256 read sentences per speaker. Each speaker read the same 32 sentences. The autosegmental-metrical analysis performed here (third author) involved transcribing this data subset for accents and phrase boundaries. In the second step, those pitch accents which featured a local f0 rise, as captured in the semi-automatic Fujisaki modelling (see 2.5), were selected and classified as belonging to either of the two basic categories, H* or L*+H.

The H* and L*+H accents which received an accent command in the automatic modelling share the presence of a rising onglide and a high target (peak), but differ in terms of alignment of both. With respect to the location of the rise onset, the H* accent was found to have two main variants, while L*+H had one (illustrated in Figure 2). Nearly all of the L*+H accents were the classic type of rise, with L stably located in the stressed syllable (type (a) of L*+H in Figure 2). In this data subset, only one L*+H did not conform to the classic pattern, and had L aligned later, i.e. in the first post-accentual syllable (type (b) of L*+H in Figure 2). The H* accents typically had the peak in the accented syllable, and the rising onglide starting in the accented syllable (type (a) of H* in Figure 2), or in the pre-accentual syllable (type (b) of H* in Figure 2). Additionally, a rare variant of H* (not illustrated in Figure 2) reminiscent of the Greek rises noted by Arvaniti, Ladd and Mennen (1998) was observed. It featured an early L (in the pre-accentual syllable, much like variant (b) of H* in Figure 2) and a late H (in the post-accentual syllable, quite like type (b) of L*+H in Figure 2); this pattern was encountered in only seven cases.

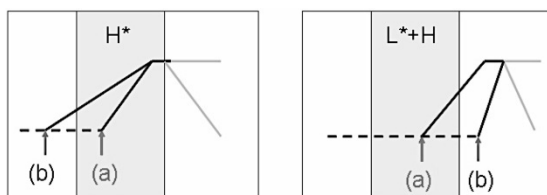


Figure 2: A schematic representation of the variants of the H* and L*+H accents in the present study's data. Points (a) and (b) indicate the L inflections of the rising onglide relative to the accented syllable, whose onset and offset are indicated by the vertical solid lines (grey box).

Bearing in mind that the accent command time-points in the semi-automatic modelling are linked to the 'current' syllable, one can infer that T_i will correctly detect variants (a) of H* and L*+H (Figure 2), whose onset of the rising onglide coincides with the 'current' syllable. The early rise onset realisations of H* (variant (b) of H* in Figure 2), however, are not currently captured by the semi-automatic match. For this reason, an additional label ('next') was created to account for those H* accents where T_i did not coincide with the location of the metrically strong (i.e., accented) syllable. 'Next' includes those cases where the accented syllable follows the occurrence of the rising onglide, hence variant (b) of H* in Figure 2, and the rare variant of H*. The frequency of the occurrence of the H*, L*+H and 'next' realisations in the ToBI annotated data subset will be covered in Section 3.3.

2.7 Statistical analysis

All data were analysed using R (R Core Team 2013) and the R packages *lme4* (Bates and Maechler 2009) and *languageR* (Baayen 2008, 2009) as well as JMP (2010). If not indicated otherwise, we analysed data using linear mixed effect models (LMEs). Normality was checked by visual inspection of quantile plots. *Speaker* and *sentence* were treated as random effects, *speaking style* (if required) as a fixed effect. Effects were tested by model comparison between a full model in which the factor in question is entered as either a fixed or a random effect (R code example: `model_full = lmer(dependent_variable ~ fixed_factor + (1|random_factor1) + (1|random_factor2), data=data)`) and a reduced model in which the factor in question is excluded (R code: `model_reduced = lmer(dependent_variable ~ 1 + (1|random_factor1) + (1|random_factor2), data = data)`). p-values were obtained by comparing the results from the two models using standard likelihood ratio tests (R code: `anova(model_full, model_reduced)`). For the assessment of the relative goodness of fit, we indicate AIC (Akaike Information Criterion) values that decrease with goodness of fit (Kliegl, Wei, Dambacher, Yan and Zhoug 2011). We assumed an α level of .05.

3. Results

The results are presented as follows. Section 3.1 shows the results of pairwise correlations for all Fujisaki f0 features: given that it is a basic tenet of FVC to accumulate as many characteristics of the speech signal as possible (Nolan 1983; Rose and Morrison 2009), we first present the results of pairwise correlations that were calculated for all derived Fujisaki model f0 features. Ideally, these parameters are not correlated, as this would reduce the number of parameters that could be used for the forensic analysis of a speaker's voice. Section 3.2 addresses research question (1) *Are there between-speaker differences in Fujisaki model f0 features?* based on data of the read corpus (speakers' 16 previously self-produced sentences + 240 sentences from their peers; all speakers read the same sentences). This section is followed by the complementary ToBI analysis for two speakers (Section 3.3). Based on a subset of spontaneous speech (those 16 sentences from the speakers' own interviews, therefore differing between speakers) and read speech (the 16 read counterparts per speaker, analogously differing between speakers), we then address research question (2) *Are these possible between-speaker differences affected by speaking style?* in Section 3.4. Research question (3) *What are the possible implications of (1) and (2) for forensic voice comparison?* is discussed in Section 4.

3.1 Pairwise correlations of Fujisaki model f0 features

To examine the degree to which the Fujisaki model f0 features explain each other, we calculated pairwise correlations. We also included articulation rate in the correlations because previous studies have shown its effect on declination (Fujisaki and Hirose 1982). Articulation rate is defined as the average syllable duration per IP:

Equation 2: Articulation rate

$$\text{syllDur_avg} = D_{\text{syll}} / N_{\text{syll}}$$

D_{syll} = sum of all syllable durations in IP (excluding pauses)

N_{syll} = number of syllables in IP

We report only correlations where $r > .2$. For the f0 temporal feature *t1abs*, only syllables that feature an f0 rise as captured by the Fujisaki model are considered, for *t2abs* only syllables that feature an f0 fall as captured by the Fujisaki model are considered; see Table 1.

Table 1: Summary of the most relevant pairwise correlations

Fujisaki model f0 feature	Result
Ap*PC dur	$r(6128) = .27, p < .0001$
t1abs*syllable duration	$r(9401) = .53, p < .0001$
t1abs*articulation rate	$r(9401) = .22, p < .0001$
Syllable duration*articulation rate	$r(53610) = .32, p < .0001$

Most of the obtained correlations are relatively low. The results did reveal a moderate correlation of *t1abs*syllable duration* ($r = .53$), however. A visual inspection of the data revealed that the longer the syllable, the later the local f0 rise. To control for syllable duration effects we calculated syllable duration-normalised *t1abs* values for the statistics to follow. This new parameter, which we call *t1rel*, expresses *t1abs* as a percentage of the duration of the syllable in which the rise onset (i.e., T_1) takes place (e.g., a value of 2% means that the rise takes place at 2% of the duration of the syllable, see Equation 3). See Figure 1 for an illustration of *t1rel*.

Equation 3: Syllable duration-normalised *t1abs* value *t1rel*

$$t1rel = (t1abs/syllDur) \times 100$$

t1abs = T_1 (starting point of f0 rise)-syllable onset

syllDur = syllable duration

3.2 Between-speaker differences in Fujisaki model f0 features in read speech

To test for speaker-specific f0 patterns, we processed LMEs for each of the six Fujisaki model f0 features (see Table 2).

Table 2: Summary of the statistics for the Fujisaki model f0 features on read speech.

Fujisaki model f0 feature	Result
Ap	$p < .0001, AIC = -2433$
PC dur	$p < .0001, AIC = 12540$
Aa	$p < .0001, AIC = -23375$
AC dur	$p < .0001, AIC = -10118$
t1rel	$p < .0001, AIC = 87417$
t2abs	$p < .0001, AIC = -17067$

All comparisons between full and reduced models showed a significant difference and all full models exhibited an increased goodness of fit, i.e., the results revealed between-speaker variation at every investigated f0 level.

3.3 ToBI analysis

The two speakers for whom the autosegmental-metrical analysis was conducted revealed significant differences in *t1rel* in the read corpus ($t(159) = 3.5$, $p = .0005$): speaker 12m showed comparatively late rises ($M = 10$, $SD = 7$), while speaker 07m exhibited relatively early rises ($M = 7$, $SD = 6$). These differences are also in agreement with the differing distribution of ToBI accents, as shown in the mosaic plot in Figure 3. The results are based on the annotated subset of identical 32 sentences of the 256 read sentences (cf. Section 2.6) for the two speakers.

Figure 3 reveals that the relative proportions of the ToBI accent categories used differ between the speakers. Speaker 07m prefers H^* accents (68%), and uses L^*+H accents less frequently (16%). Speaker 12m, on the other hand, shows a clear preference for L^*+H accents (65%) and uses H^* accents less frequently (29%). To corroborate these descriptive statistics inferentially, we processed a Chi-squared test of independence. The test confirmed a significant difference in the relative proportions of ToBI accents between the two speakers ($\chi^2(3, 131) = 34$, $p < .0001$). The relatively low proportion of ‘next’ (16% for speaker 07m and 6% for speaker 12m) indicates that the early rise onset realisations of H^* occur much less frequently than the variant of H^* with the rise onset in the accented syllable (see variants (a) and (b) of H^* in Section 2.6).

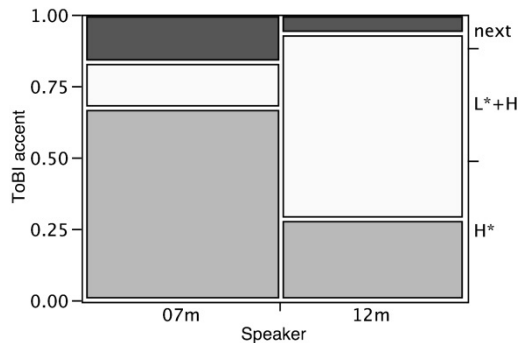


Figure 3: Mosaic plot of the different types of ToBI accents found in the same 32 read sentences per speaker

3.4 Comparison between read and spontaneous speech

To test the effects of *speaker* and *speaking style*, we processed LMEs in the form described in Section 2.7. To test for an interaction of *speaking style* and *speaker*, we calculated an LME and performed standard likelihood ratio tests comparing the model with *speaking style*speaker* as an interaction term and *sentence* as

random effect (full model) to a model with *speaking style* and *speaker* as fixed effects and *sentence* as random effect. For *t1rel*, only syllables that feature an *f0* rise as captured by the Fujisaki model are considered; for *t2abs* only syllables that feature an *f0* fall as captured by the Fujisaki model are considered. Table 3 summarises the results obtained.

Table 3: Summary of the statistics for the Fujisaki model *f0* features tested on a subset of read and spontaneous speech

Fujisaki model <i>f0</i> feature	speaker	speaking style	speaker*speaking style
<i>Ap</i>	$p < .0001$, AIC = -164	$p < .0001$, AIC = -221	$p < .0001$, AIC = -340
<i>PCdur</i>	n.s.	$p < .0001$, AIC = 153	n.s.
<i>Aa</i>	$p < .0001$, AIC = -2303	n.s.	$p < .0001$, AIC = -2348
<i>ACdur</i>	n.s.	$p < .0001$, AIC = -1327	n.s.
<i>t1rel</i>	n.s.	n.s.	n.s.
<i>t2abs</i>	$p = .0024$, AIC = -1892	$p < .0002$, AIC = -1903	n.s.

Ap: Figure 4 shows the boxplots of the speakers’ global declination parameter *Ap* in spontaneous speech (white) and read speech (grey). The model with *speaker* as random effect provided an improved goodness of fit, i.e. between-speaker variation was significant. Results further indicated a significant difference in declination across the two speaking styles, with read speech showing steeper declination ($M = .6$, $SD = .29$) than spontaneous speech ($M = .49$, $SD = .25$). There was a significant interaction of *speaking style*speaker*. To test for *speaker* as a simple effect, we computed one LME on the spontaneous sen-

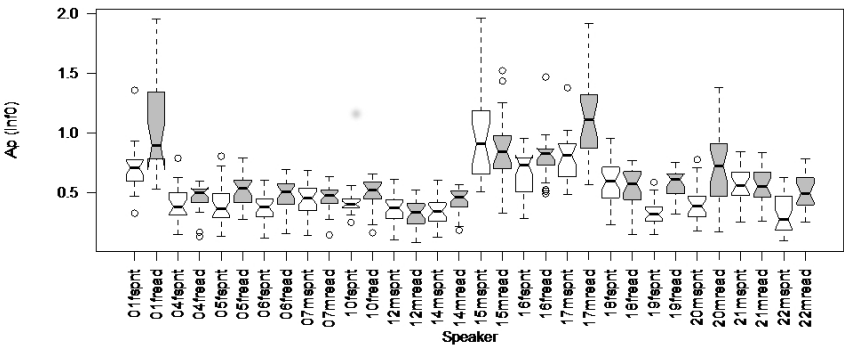


Figure 4: Boxplots of the 16 speakers’ global declination parameter *Ap*: spontaneous speech (white) and read speech (grey)

tences and one LME on the read sentences. Both LMEs showed significant effects of *speaker* (Bonferroni adjusted for *speaking style*, $\alpha = 0.025$; spontaneous: $p < .0001$, AIC = -221; read: $p < .0001$, AIC = -44). To test for the simple effect of *speaking style*, we again computed LMEs. For 9 out of 16 speakers, there were no significant differences between the two speaking styles (Bonferroni adjusted for *speaker*, $\alpha = 0.0031$). For 7 speakers, there was a significant difference (speaker 01f: $p = .001$, AIC = 33; speaker 05f: $p = .002$, AIC = -53; speaker 06f: $p = .001$, AIC = -58; speaker 17m: $p < .0001$, AIC = 16; speaker 19f: $p < .0001$, AIC = -82; speaker 20m: $p < .0001$, AIC = 8; speaker 22m: $p < .0001$, AIC = -43).

PC dur: The model with *speaker* as random effect did not provide an improved goodness of fit. The results indicated a significant difference in phrase duration across the two speaking styles, with read speech showing longer IPs ($M = 2.1$, $SD = .7$) than spontaneous speech ($M = 1.7$, $SD = .58$). There was no interaction of *speaking style* and *speaker*.

Aa: Figure 5 shows the boxplots of the speakers' local f_0 excursions in spontaneous speech (white) and read speech (grey). The model with *speaker* as a random effect provided an improved goodness of fit, i.e. between-speaker variation was significant. The results further revealed that excursions of f_0 do not differ between the two speaking styles. There was a significant interaction of *speaking style*speaker*. To test for speaker as a simple effect, we computed one LME on the spontaneous sentences and one LME on the read sentences. Both LMEs showed significant effects of *speaker* (Bonferroni adjusted for *speaking style*, $\alpha = 0.025$; spontaneous: $p < .0001$, AIC = -914; read: $p < .0001$, AIC = -1473).

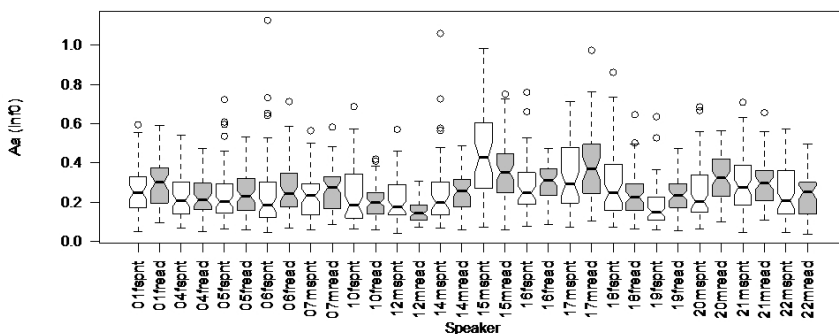


Figure 5: Boxplots of the 16 speakers' local f_0 excursion values: spontaneous speech (white) and read speech (grey)

To test for the simple effect of *speaking style*, we again computed LMEs. For 13 out of 16 speakers, there were no significant differences between the two speaking styles (Bonferroni adjusted for *speaker*, $\alpha = 0.0031$). For 3 speakers, there was a significant difference (speaker 12m: $p < .0001$, $AIC = -274$; speaker 19f: $p < .0001$, $AIC = -227$; speaker 20m: $p = .0013$, $AIC = -170$). This is taken as support for the subordinate role of *speaking style* in within-speaker variation for this Fujisaki model f0 feature.

AC dur: The model with *speaker* as random effect did not provide an improved goodness of fit, i.e. between-speaker variation was not distinct. The results did indicate, however, a significant difference in AC duration across the two speaking styles, with read speech showing slightly longer accent durations ($M = .38$, $SD = .17$) than spontaneous speech ($M = .35$, $SD = .17$). There was no significant interaction of *speaking style*speaker*.

t1rel: The model with *speaker* as random effect did not provide an improved goodness of fit, nor was there a significant effect of *speaking style* or interaction of *speaking style*speaker*.

t2abs: Figure 6 shows the boxplots of the speakers' *t2abs* values in spontaneous speech (white) and read speech (grey). The point 0.0 on the y-axis in Figure 6 denotes syllable offset. The lower the *t2abs* value, the earlier the f0 fall relative to syllable offset. The boxplots reveal between-speaker differences: speaker 10f, for instance, initiates f0 falls comparatively late relative to syllable offsets in both speaking styles (spontaneous: $M = -.08$, $SD = .05$, read: $M = -.08$, $SD = .08$) while speaker 22m shows relatively early f0 falls in both speaking styles

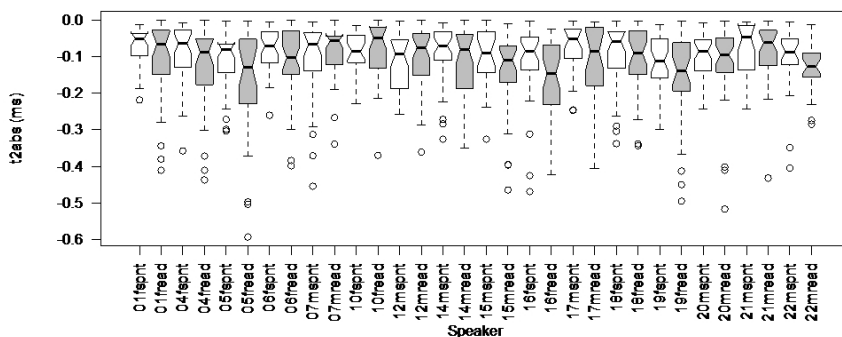


Figure 6: Boxplots of the 16 speakers' *t2abs* values: spontaneous speech (white) and read speech (grey)

(spontaneous: $M = -.1$, $SD = .09$; read: $M = -.12$, $SD = .07$). The model with *speaker* as random effect provided an improved goodness of fit, i.e. between-speaker variation was significant (see Table 3). The results further revealed that read speech overall exhibits earlier falls ($M = -.11$, $SD = .1$) than spontaneous speech ($M = .097$, $SD = .082$). There was no significant interaction of *speaking style*speaker*.

4. Discussion

In the present study we reported evidence that a homogeneous group of speakers strongly varies with respect to a number of Fujisaki model f0 features. We showed that a selection of these features remained relatively stable when speaking style varied.

Pairwise correlations of the Fujisaki model f0 features showed interesting tendencies. A spot check of the data revealed that IPs with steeper declination trends tend to be longer, hence the correlation of *Ap (declination)*PC dur* (see Table 1). The phrase component rises quickly and slowly decreases to the baseline F_b . This lowest point is reached more rapidly the flatter the declination (see Fujisaki and Hirose 1982; Leemann 2012). Table 1 further revealed a moderate correlation of *t1abs*syllable duration*: a visual inspection corroborated the finding that the longer the syllable, the later the rise. The later rises may be facilitated by the increased duration of the syllable: even if T_1 occurs late, a pitch peak realisation within a long target segment is still possible. If syllables are shorter, T_1 occurs earlier so as to ensure peak realisation in the target segment.

4.1 Are there between-speaker differences in Fujisaki model f0 features?

The results indicated that for the read speech corpus (256 sentences per person: 16 previously self-produced sentences + 240 sentences from their peers), speakers significantly differed from one another in every Fujisaki model f0 feature investigated: declination, PC duration, local excursion of f0, duration of local excursion of f0, as well as temporal alignment of f0 inflections relative to syllable onsets and offsets. The present study is a first account of speaker-individuality in these f0 features. The additional autosegmental-metrical analyses of the speakers that showed the greatest differences in *t1rel* revealed that each of these speakers shows a preference for a particular accent type. These findings complement previous research that reported speaker-individuality on other f0 levels: f0 means and standard deviations (see Braun 1992; Gold and French 2011; Jessen et al. 2005; Künzel et al. 1995; Künzel 2000; Labutin et al. 2007; Nolan 2002; Zellner Keller 2006), f0 differences between the onset and

end of the first rise, durations of the second rise in declaratives, duration of final lowering (see Kraayeveld 1997, terminology as used by Kraayeveld), f0 range and rate of fall (Abberton and Fourcin 1978, terminology as used by Abberton and Fourcin), and the point in time at which the f0 contour reaches the f0 halfway between high and low targets (Nolan 2002).

4.2 Are these possible between-speaker differences affected by speaking style?

The results revealed significant effects for the factor *speaker* across both *speaking styles* for declination, local f0 excursions and the temporal parameter *t2abs*. The one Fujisaki f0 feature that seems to be most robust to speaking style variability is the local f0 excursion. How speakers modulate local f0 for prominence marking appears to be speaker-individual and unaffected by changes in speaking style. For declination and *t2abs* there was an effect of *speaking style*. From a linguistic point of view, read speech showed steeper declination, earlier falls and longer AC durations. The finding that declination is more distinct in read as opposed to spontaneous speech has been previously reported in the literature (see Umeda 1982; Vaissière 1983).

4.3 What are the possible implications of (1) and (2) for forensic voice comparison?

Given the basic tenet of FVC to accumulate as many characteristics of the speech signal as possible (see Nolan 1983; Rose and Morrison 2009), the present contribution has shown that Fujisaki model f0 features obtained through semi-automatic analysis may further help maximise between-speaker differences. In FVC, incorporation of the Fujisaki model f0 features may be advantageous compared to an analysis of features such as f0 means and standard deviations for the following reason: while an impostor may be able to modify such global f0 features of his/her voice (e.g., average f0, see Künnzel 2000), it is questionable whether s/he can easily disguise f0 features such as local excursions of f0 or temporal f0 features such as those encoded in *t2abs*, for example (see Atal 1972). The fact that particularly local f0 excursions did not differ between the two speaking styles is further useful for FVC, since casework and suspect material may differ in speaking style. Our results further indicated that speakers seem to have preferences for certain phonological accent types as measured by the complementary ToBI annotation, which additionally contributes to speaker-individuality that could be leveraged in FVC.

Another result that may have implications for FVC is what we found for articulation rate and its relation to syllable duration. When we correlated the durations of syllables in which a local f0 rise takes place with the average syllable durations of the phrase – which we associate with articulation rate (see

Equation 2) – we find only a weak correlation ($r = .32$). We take this to mean that syllable durations in which f_0 rises occur stand largely independent of overall articulation rate. In other words, if a speaker wants to manipulate speech rate – which frequently occurs in forensic cases (see Künzel 2000) – then it is unlikely that s/he will be able to manipulate rate in syllables that include a local f_0 rise. The syllables that feature an f_0 rise seem to have to be held constant in duration and are thus particularly interesting for FVC.

In Section 1 it was mentioned that a semi-automatic as opposed to a manual f_0 parameterisation is critical, since this increases the chances that the obtained between-speaker differences are uniformly detected by the algorithm and are not a by-product of the modelling approach which can differ between modellers. This semi-automatic modelling comes at a trade-off, however: since the modelling is performed purely on the basis of the acoustic signal, subsequent linguistic interpretations may be somewhat limited. The current study's definition of an IP, for example, which we conceived of as the interval duration between two consecutive T_0 s (see Section 2.4), does not directly translate to a linguistically motivated definition of an IP, which would state that at least one accent is present which serves as the tonal anchor point for the intonation contour (see Gilles 2005). Moreover, the IPs obtained through our semi-automatic f_0 modelling may not fully match the linguistically motivated major or minor phrases as applied in the ToBI framework (Silverman et al. 1992), for example. In future studies, a more linguistically driven parameterisation would complement and quite possibly enhance the speaker-discriminating potential of Fujisaki model f_0 features.

In a linguistically constrained approach, phrase and accent commands are matched to the actual phrases and accents. Additional, linguistically unjustified commands are disregarded (i.e., deleted) even if the resulting fit is mathematically less accurate. In the purely automated match – as applied in the current study – extra (non-phrase-initial) phrase commands may be used if the shape of the f_0 contour requires such. Also, accent commands are assigned to any local f_0 movement, thus modelling not only accents and boundary tones, but also non-accent-lending and micro-prosodic f_0 movements, if such have not been previously removed. Between-speaker differences would quite plausibly be even more distinct for T_I -derived values, if T_I was linked to metrically stressed syllables (instead of 'current' syllables, as is done by the automatic Fujisaki parameterisation applied in the present study): the H^* category would most likely show negative to mildly positive T_I values, L^*+H would most likely show higher positive T_I values if the modelling were to be done manually. On the other hand, the results of the current study's autosegmental-metrical analysis were added to the Fujisaki model T_I results,

and not vice versa. We acknowledge the possibility that if the autosegmental-metrical analysis was the starting point of analysis, there could be cases of missing accent commands on metrically strong syllables, although the automatic match may be more likely to overestimate rather than to underestimate the number of accent commands in the modelling.

5. Conclusion

Largely unbiased by the selectivity of a human modeller, the present study was a first attempt at finding speaker-specific f_0 parameters with the use of the Fujisaki intonation model. The results revealed significant between-speaker differences in all tested Fujisaki model f_0 features: f_0 declination, PC duration, local excursions of f_0 , duration of f_0 excursions of f_0 , as well as temporal alignment of f_0 inflections relative to syllable onsets and offsets. Further analyses revealed that a number of these features – declination, local f_0 excursion and $t2abs$ – showed significant effects across both speaking styles and that speakers seem to have individual preferences for the use of particular accent types. Given this evidence for speaker-individuality, we argue that the Fujisaki model f_0 features could complement the existing, largely coarse, f_0 statistics that are currently used in FVC.

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References

- Abberton, E. and Fourcin A. J. (1978) Intonation and speaker identification. *Language and Speech* 21(4): 305–318.
- Adriaens, L. M. H. (1991) Ein Modell deutscher Intonation. PhD thesis, University of Technology, Eindhoven.

- Arvaniti, A., Ladd, D. R. and Mennen, I. (1998) Stability of tonal alignment: the case of Greek prenuclear accents. *Journal of Phonetics* 26: 3–25.
<http://dx.doi.org/10.1006/jpho.1997.0063>
- Atal, B. S. (1972) Automatic speaker recognition based on pitch contours. *Journal of the Acoustical Society of America* 52(6): 1687–1697. <http://dx.doi.org/10.1121/1.1913303>
- Baayen, R. H. (2008) *Analyzing Linguistic Data: A Practical Introduction to Statistics Using R*. Cambridge: Cambridge University Press.
<http://dx.doi.org/10.1017/CBO9780511801686>
- Baayen, R. H. (2009) *languageR: Data Sets and Functions with 'Analyzing Linguistic Data: A Practical Introduction to Statistics using R'*. R package version 0.955.
- Bates, D. M. and Maechler, M. (2009) *lme4. Linear Mixed-Effects Models Using S4 Classes*. R package version 0.999375-32.
- Baumann, S. (2006) Information structure and prosody: linguistic categories for spoken language annotation. In S. Sudhoff et al. (eds) *Methods in Empirical Prosody Research* 153–180. Berlin: de Gruyter.
- Boersma, P. (1993) Accurate short-term analysis of the fundamental frequency and the harmonics-to-noise ratio of sampled sound. *Proceedings of the Institute of Phonetic Sciences, University of Amsterdam* 17: 97–100.
- Boersma, P. and Weenink, D. (2014) *Praat – Doing Phonetics by Computer*.
<http://www.fon.hum.uva.nl/praat/>
- Bolinger, D. (1989) *Intonation and its Uses: Melody in Grammar and Discourse*. Stanford, CA: Stanford University Press.
- Botinis, A., Granström, B. and Möbius, B. (2001) Developments and paradigms in intonation research. *Speech Communication* 33: 263–296. [http://dx.doi.org/10.1016/S0167-6393\(00\)00060-1](http://dx.doi.org/10.1016/S0167-6393(00)00060-1)
- Braun, A. (1992) Zur Bedeutung des Merkmals mittlere 'Sprechstimmlage' in der forensischen Sprechererkennung. In H. R. Dingeldein (ed.) *Festschrift für J. Göschel* 1–26. Marburg: Universitätsbibliothek.
- Braun, A. (1995) Fundamental frequency – How speaker-specific is it? In A. Braun and J. P. Köster (eds) *Studies in Forensic Phonetics* 9–23. Beiträge zur Phonetik und Linguistik 64 Trier: Wissenschaftlicher Verlag.
- Duez, D. (1982) Silent and non-silent pauses in three speech styles. *Language and Speech* 25: 11–28.
- Fleischer, J. and Schmid, S. (2006) Zurich German. *Journal of the International Phonetic Association* 25(2): 243–253. <http://dx.doi.org/10.1017/S0025100306002441>
- Fujisaki, H. (1981) Dynamic characteristics of voice fundamental frequency in speech and singing. *Acoustical Analysis and Physiological Interpretations, Quarterly Progress and Status Report, Department for Speech, Music and Hearing, KTH Stockholm*: 1–20.

- Fujisaki, H. (1987) A note on the physiological and physical basis for the phrase and accent components in the voice fundamental frequency contour. *Annual Bulletin, Research Institute for Logopaedics and Phoniatrics, Faculty of Medicine, University of Tokyo* 21: 65–75.
- Fujisaki, H. (1992a) The role of quantitative modeling in the study of intonation. *Proceedings of the International Symposium on Japanese Prosody*: 163–174.
- Fujisaki, H. (1992b) Modeling the process of fundamental frequency contour generation. In Y. Tohkura, E. Vatikiotis-Bateson and Y. Sagisaka (eds) *Speech Perception, Production, and Linguistic Structure* 313–326. London: IOS Press.
- Fujisaki, H. (2004) Prosody, information, and modeling – with emphasis on tonal features of speech. *Proceedings of Speech Prosody 2004, Nara, Japan*: 1–10.
- Fujisaki, H. and Hirose, K. (1982) Modeling the dynamic characteristics of voice fundamental frequency with applications to analysis and synthesis of intonation. *Preprints of the Working Group on Intonation, 13th International Congress of Linguistics, Tokyo*: 57–70.
- Fujisaki, H., Hirose, K. and Ohta, K. (1979) Acoustic features of the fundamental frequency contours of declarative sentences in Japanese. *Annual Bulletin, Research Institute for Logopaedics and Phoniatrics, Faculty of Medicine, University of Tokyo* 13: 163–172.
- Fujisaki, H., and Nagashima, S. (1969) A model for the synthesis of pitch contours of connected speech. *Annual Report of the Engineering Research Institute, Faculty of Engineering, University of Tokyo*.
- Fujisaki, H., Tatsumi, M. and Higuchi, N. (1980) Analysis of pitch control in singing. *Annual Bulletin, Research Institute of Logopedics and Phoniatrics, Faculty of Medicine, University of Tokyo* 14: 101–111.
- Gfroerer, S. and Wagner, I. (1995) Fundamental frequency in forensic speech samples. In A. Braun and J.P. Köster (eds) *Studies in Forensic Phonetics* 41–48. Beiträge zur Phonetik und Linguistik 64. Trier: Wissenschaftlicher Verlag.
- Gilles, P. (2005) *Regionale Prosodie im Deutschen: Variabilität in der Intonation von Abschluss und Weiterweisung*. Berlin: de Gruyter.
- Gold, E. and French, P. (2011) International practices in forensic speaker comparison. *International Journal of Speech, Language, and the Law* 18(2): 293–307.
- Goldsmith, J. (1976) *Autosegmental and Metrical Phonology*. New York: Garland.
- Hall, T. (2011) *Phonologie: Eine Einführung*. Berlin: de Gruyter.
- Hirst, D. J. and Di Cristo, A. (1998) A survey of intonation systems. In D. J. Hirst and A. Di Cristo (eds) *Intonation Systems: A Survey of Twenty Languages* 1–44. Cambridge: Cambridge University Press.
- Hollien, H. and Jackson, B. (1973) Normative data on the speaking fundamental frequency characteristics of young adult males. *Journal of Phonetics* 1: 117–120.

- Jessen, M. (2009) Forensic phonetics and the influence of speaking style on global measures of fundamental frequency. In G. Grewendorf and M. Rathert (eds) *Formal Linguistics and Law* 115–139. Berlin: Mouton de Gruyter.
- Jessen, M., Köster, O. and Gfroerer, S. (2005) Influence of vocal effort on average and variability of fundamental frequency. *Journal of Speech, Language and the Law* 12(2): 174–213.
- JMP (2010) Version 9.0, Cary NY, SAS Institute Inc.
- Kehrein, R. (2002) *Prosodie und Emotionen*. Tübingen: Niemeyer.
- Kliegl, P., Wei, P., Dambacher, M., Yan, M. and Zhou, X. (2011) Experimental effects and individual differences in linear mixed models: estimating the relationship between spatial, object, and attraction effects in visual attention. *Frontiers in Psychology* 1(238): 1–12.
- Kraayeveld, H. (1997) Idiosyncrasy in prosody. Speaker and speaker group identification in Dutch using melodic and temporal information. Doctoral thesis, Katholieke Universiteit Nijmegen.
- Künzel, H., (2000) Effects of voice disguise on speaking fundamental frequency. *Forensic Linguistics* 7(2): 149–179.
- Künzel, H., Masthoff, H. R. and Köster, J. P. (1995) The relation between speech tempo, loudness, and fundamental frequency: an important issue in forensic speaker recognition. *Science and Justice* 35(4): 291–295. [http://dx.doi.org/10.1016/S1355-0306\(95\)72683-1](http://dx.doi.org/10.1016/S1355-0306(95)72683-1)
- Labutin, P., Koval, S. and Raev, A. (2007) Speaker identification based on the statistical analysis of f0. *Proceedings of IAFPA 2007, Plymouth, UK*.
- Ladd, D. R. (1996) *Intonational Phonology*. Cambridge: Cambridge University Press.
- Leemann, A. (2012) *Swiss German Intonation Patterns*. Amsterdam and New York: Benjamins. <http://dx.doi.org/10.1075/silv.10>
- Lindh, J. and Eriksson, A. (2007) Robustness of long time measures of fundamental frequency. *Proceedings of Interspeech 2007, Antwerp, Belgium*: 2025–2028.
- Mixdorff, H. (1998) Intonation patterns of German – model-based quantitative analysis and synthesis of F0-contours. PhD thesis, TU Dresden.
- Mixdorff, H. (2000) A novel approach to the fully automatic extraction of Fujisaki model parameters. *Proceedings of ICASSP 2000, Istanbul, Turkey*: 1281–1284.
- Mixdorff, H. (2004) Quantitative tone and intonation modeling across languages. *Proceedings of TAL 2004, Beijing, China*: 137–142.
- Mixdorff, H. (2009) *FujiParaEditor. Fujisaki model Parameter Extraction Environment*. <http://public.beuth-hochschule.de/~mixdorff/thesis/fujisaki.html> (retrieved on 20 August 2014).
- Mixdorff, H. and Fujisaki, H. (2000) A quantitative description of German prosody offering symbolic labels as a by-product. *Proceedings of ICSLP 2000, Beijing, China*: 98–101.
- Möbius, B. (1993) *Ein quantitatives Modell der deutschen Intonation: Analyse und*

Synthese von Grundfrequenzverläufen. Tübingen: Niemeyer.
<http://dx.doi.org/10.1515/9783111355870>

- Murray, I. R. and Arnott, J. L. (1993) Toward the simulation of emotion in synthetic speech: a review of the literature on human vocal emotion. *Journal of the Acoustic Society of America* 93: 1097–1108. <http://dx.doi.org/10.1121/1.405558>
- Nolan, F. (1983) *The Phonetic Bases of Speaker Recognition*. Cambridge: Cambridge University Press.
- Nolan, F. (2002) Intonation in speaker identification: an experiment on pitch alignment features. *International Journal of Speech, Language and the Law* 9(1): 1–21.
- O'Reilly, M. (2013) Sentence mode, alignment and focus in the intonation of Cois Fharraige, Inis Mór and Gaoth Dobhair – A dual approach. PhD thesis, Trinity College Dublin.
- O'Reilly, M., Ni Chasaide, A. and Gobl, C. (2008) Cross-dialect Irish prosody: linguistic constraints on Fujisaki modeling. *Proceedings of Interspeech 2008, Brisbane, Australia*: 886–889.
- R Core Team (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computer, Version 3.0.0, <http://www.R-project.org>
- Rose, P., and Morrison, G. S. (2009) A response to the UK Position Statement on forensic speaker comparison. *International Journal of Speech, Language and the Law* 15(1): 139–163.
- Selting, M. (1995) *Prosodie im Gespräch: Aspekte einer interaktionalen Phonologie der Konversation*. Tübingen: Niemeyer. <http://dx.doi.org/10.1515/9783110934717>
- Siebenhaar, B. (2014) Phonological and phonetic considerations for a classification of Swiss German dialects as a word language or syllable language. In J. C. Reina and R. Szczepaniak (eds) *Syllable and Word Languages* 327–345. Berlin: de Gruyter.
- Sievers, E. (1881) *Grundzüge der Phonetik*. Leipzig: Breitkopf und Hartel.
- Silverman, K., Beckman, M. E., Pitrelli, J., Ostendorf, M., Wightman, C., Price, P., Pierrehumbert, J. and Hirschberg, J. (1992) TOBI: A standard for labeling English prosody. *Proceedings of the International Conference on Spoken Language Processing* 2: 867–870.
- Taylor, P. (1994) *A Phonetic Model of Intonation in English*. Bloomington, IN: Indiana University Linguistics Club Publications.
- Umeda, N. (1982) 'F0 declination' is situation dependent. *Journal of Phonetics* 10: 279–290.
- Vaissière, J. (1983) Language-independent prosodic features. In A. Cutler and D. R. Ladd (eds), *Prosody: Models and Measurement* 53–66. New York: Springer.
- Zellner Keller, B. (2006) F0 and intensity distributions of Marsec speakers: types of speaker prosody. In M. Faundez-Zanuy et al. (eds) *Nonlinear Analyses and Algorithms for Speech Processing. Lecture Notes in Computer Science* 116–124. Berlin: Springer. http://dx.doi.org/10.1007/11613107_10

Appendix

The first 20 of the 256 transcribed sentences of the read corpus:

- 1 So s Typische was sich d Lüüt vorscheled isch Kurator.
- 2 Ich han Freiziit.
- 3 Ich han käi äigeni Band.
- 4 Ich bin wäge Spraachwüesseschaft dänn usegheit.
- 5 Das han i cool gfunde.
- 6 Mini Mueter isch ä no nie z Wien gsi.
- 7 Dänn mues ich ä no überlegge, was mis nöie Hauptfach wird.
- 8 Ich ha jetz äifach verglichendi Spraachwüesseschafte gno.
- 9 Ich ha mich ä nie würlklich beworbe.
- 10 Wänn ich halt im Usland wär, hett ich das zmindescht mal für es Semeschter nöd.
- 11 Chasch ja nöd nöime andersch go studiere mit Erasmus.
- 12 Si liit det am Bode.
- 13 Zwar isch das Ganze im ne fiktive Königriich.
- 14 Ich wäis nöd werum si so abglänkt isch.
- 15 Säge mer emaal ich fahr uf Oerlike.
- 16 Mit em Zug sälber zwänzg Minute.
- 17 Wil's äifach di zwäiti Spraach isch.
- 18 Das git's äigentlich i käinere andere Spraach.
- 19 Tschechisch han i käi Phonetik ghaa.
- 20 Ich glaub mi händ so chli ä drüber gredt ghaa.